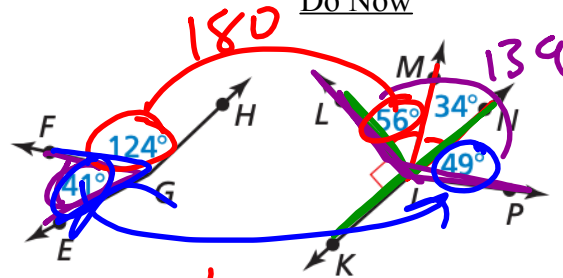


Students will describe, identify, and use the properties of angle pairs to solve problems.

Do Now



1. Name a pair of adjacent complementary angles.
 $\angle LJM$ and $\angle NJM$ (add to 90)
2. Name a pair of adjacent supplementary angles.
 $\angle LJM$ and $\angle LJK$ (180)
3. Name a pair of nonadjacent supplementary angles.
 $\angle EGF$ and $\angle LJP$
4. Name a pair of nonadjacent complementary angles.

$\angle FGE$ and $\angle NGP$

Do Next

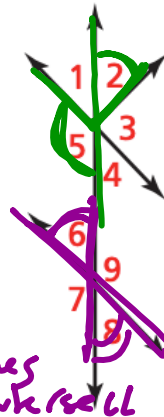
- 5) a) Identify all the linear pairs that include $\angle 1$. $\angle 1$ and $\angle 5$

- b) Identify all the linear pairs that include $\angle 7$. $\angle 7$ and $\angle 8$
 $\angle 7$ and $\angle 6$

- c) Are $\angle 6$ and $\angle 8$ vertical angles?
Explain your reasoning. Yes,

- d) Are $\angle 2$ and $\angle 5$ vertical angles?
Explain your reasoning. No

because they are not formed by 2 intersecting lines.



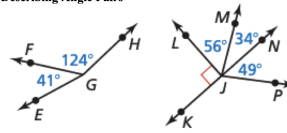
Questions on Homework?

Work in This Order

- 1) Complete House Flooring Project
- 2) Complete Incomplete Unit 1 Geometry Assignments
- 3) Complete Assignment 1.9b (handed out today)

Assignment 1.9b: Describing Angle Pairs

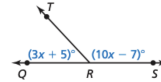
1. Name a pair of the indicated angles.
 - a. Adjacent Complementary Angles
 - b. Adjacent Supplementary Angles
 - c. Nonadjacent Complementary Angles
 - d. Nonadjacent Supplementary Angles



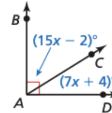
Use the given information to find the measure of the unknown angle.

2. $\angle 3$ is a complement of $\angle 4$, and $m\angle 3 = 46^\circ$
3. $\angle 7$ is a supplement of $\angle 8$, and $m\angle 7 = 109^\circ$

4. Find the measure of each angle.
 - a. Set up an equation that represents the relationship in the diagram.
 - b. Solve for x
 - c. Find $m\angle QRT$ and $m\angle TRS$



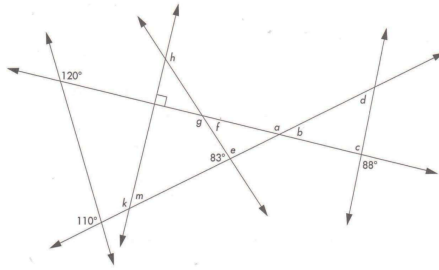
5. Find the measure of each angle.
 - a. Set up an equation that represents the relationship in the diagram.
 - b. Solve for x
 - c. Find $m\angle BAC$ and $m\angle CAD$



6. Use the diagram to answer the following questions.
 - a. Identify all the linear pairs that include $\angle 5$.
 - b. Identify all the linear pairs that include $\angle 6$.
 - c. Are $\angle 6$ and $\angle 8$ vertical angles?
 - d. Are $\angle 2$ and $\angle 5$ vertical angles?
7. Two angles form a linear pair. The measure of the smaller angle is 102° less than the measure of the larger angle. Find the measure of each angle.
8. Two angles are complementary. The measure of the larger angle is 5 times the measure of the smaller angle. Find the measure of each angle.
9. Two angles are supplementary. The measure of the larger angle is 6° less than twice the measure of the smaller angle. Find the measure of each angle.



10. Fun fact: The angles inside of a triangle must add to 180° . Use this fact to find all of the missing angles in the image below. Show your work on the paper below when possible.



a =
b =
c =
d =
e =
f =
g =
h =

Optional Challenge

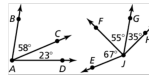
Two lines intersect at point O , forming four angles. The measure of one angle is three times the measure of an adjacent angle.

- Draw a picture of this scenario.
- Find the measures of all four angles. Explain how you know.
- A ray is drawn from O , dividing the smaller angle into two complementary angles. Is this possible? Explain.
- A ray is drawn from O , dividing the larger angle into two angles, one of which is twice the other. Find the measures of the two new angles.
- Create a different pair of intersecting lines where one angle is four times its adjacent angle. Find all four angle measures.

Assignment 1.9: Describing Pairs of Angles

In Exercises 1–3, use the figures.

1. Name a pair of adjacent complementary angles.
2. Name a pair of nonadjacent complementary angles.
3. Name a pair of nonadjacent supplementary angles.



In Exercises 4 and 5, find the angle measure.

4. $\angle 1$ is a complement of $\angle 2$, and $m\angle 2 = 36^\circ$. Find $m\angle 1$.
5. $\angle 3$ is a supplement of $\angle 4$, and $m\angle 4 = 75^\circ$. Find $m\angle 3$.

In Exercises 6 and 7, find the measure of each angle.

6. $\angle WXY$ and $\angle YXZ$ are supplementary angles, $m\angle WXY = (6x + 59)^\circ$, and $m\angle YXZ = (3x - 14)^\circ$.

a) Draw a picture matching the description.

b) Set up a relationship representing the description and picture you created.

c) Solve for x .d) Find $m\angle WXY$ and $m\angle YXZ$.

7. $\angle ABC$ and $\angle CBD$ are complementary angles, $m\angle ABC = (3x + 6)^\circ$, and $m\angle CBD = (4x - 14)^\circ$.

a) Draw a picture matching the description.

b) Set up a relationship representing the description and picture you created.

c) Solve for x .d) Find $m\angle WXY$ and $m\angle YXZ$.

Two angles are complementary.

The smaller angle is 18 less than the larger $\angle$.

Two ways to think about this:

① Small $\angle$ is x .
Then large $\angle$ is
 $x + 18$

$$x + x + 18 = 90$$

② Large $\angle$ is x .
Then small $\angle$ is
 $x - 18$

$$x + x - 18 = 90$$

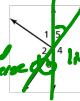
In Exercises 8–10, use the figure.

8. Identify the linear pairs that include $\angle 5$.

9. Are $\angle 3$ and $\angle 5$ vertical angles? Explain your reasoning.

y - opposite $\angle$ s formed by 2 intersecting lines.

10. Are $\angle 2$ and $\angle 4$ vertical angles? Explain your reasoning.



In Exercises 11–13, write and solve an algebraic equation to find the measure of each angle based on the given description.

11. Two angles form a linear pair. The larger angle's measure is 24° more than the measure of the smaller angle.

12. Two angles are complementary. The larger angle's measure is three times the measurement of the smaller angle.

Advice: Take their word

13. Two angles are supplementary. The smaller angle's measure is 15 less than half the measurement of the larger angle.



so small = $\frac{1}{2}x - 15$
large = x

minus

$$1x + \frac{1}{2}x - 15 = 180$$

$$1.5x - 15 = 180$$

$$+15 \quad +15$$

$$\frac{1.5x = 195}{1.5}$$

$$x = 130$$

14. The figure shows the design on an outdoor fence.

a. Name a pair of adjacent supplementary angles.

b. Name a pair of nonadjacent supplementary angles.

c. Identify the linear pairs that include $\angle 5$.

d. Find $m\angle 3$. Explain your reasoning.



Work order

1) House Floor Project / Retake /
 old assign

2) 1.9

3) 1.96

